الاسم والشهرة:----------

رقم المرشح :----------

الجمهورية اللبنانية
وزارة التربية والتعليم العالي
المديرية العامة للتعليم المهني والتقني
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الامتحانات الرسمية لشهادة الامتياز الفني
لدورة عام 2018 الاولى

الاختصاص والرمز : الالكترونيك (20210)
المادة : الأعمال التطبيقية : الكترونيك رقمي
المستندات المسموح بها : آلة حاسبة غير مبرمجة

المعدل : (2)
المدة : ساعة واحدة

Travail à faire:

Ecrire le programme en langage assembleur demandé sur le logiciel correspondant, compiler, simuler et visualiser le résultat demandé.

Taiter pratiquement un exercice parmi deux:

I- MP 6809

Ecrire un programme en assembleur 6809 pour faire clignoter ou non le PORTB, du PIA 6821, selon le niveau du pin PA0 du port A: si PA0=0 alors PORTB = 00 et si PA0=1 alors PORTB clignote (FF-00-FF-00-....) avec un retard à votre choix.

II- PIC 16F84A

Ecrire un programme qui permet de faire sortir sur le PORTB, séquentiellement, les huit octets du tableau ci-dessous avec un retard à votre choix.

Work to be executed

Write the program in required assembly language on the corresponding software, compile, simulate and visualize the required result.

Treat Praqticaly one exercise between two:

I- MP 6809 :

Write a program in assembly 6809 that flashing or no the PORTB, of the PIA 6821, according to the level of pin PA0 of the PORTA: if PA0 = 0 then PORTB = 00 and if PA0 = 1 then PORTB flashing (FF-00-FF-00-…) with a delay at your choice.

II- PIC 16F84A

Write a program that sequentially outputs at PORTB, the 8 bytes of the table below with a delay at your choice.

RB7	RB6	RB5	RB4	RB3	RB2	RB1	RB0
0	1	1	1	1	1	1	1
0	0	1	1	1	1	1	1
0	0	0	1	1	1	1	1
0	0	0	0	1	1	1	1
0	0	0	0	0	1	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	0	1
0	0	0	0	0	0	0	0

Utilisation du logiciel	6 pts	*Use of the software*	*6 pts*
Compilation correcte	4 pts	*Right compilation*	*4 pts*
Simulation correcte	4 pts	*Right simulation*	*4 pts*
Questions orales de compréhension	6 pts	*Oral questions of comprehension*	*6 pts*

ANNEXE

PIA6821:

ORA et DDRA = $8000	**CRA = $8001**
ORB et DDRB = $8002	**CRB = $8003**

Instruction set of PIC 16F84A

Table 29-1: Midrange Instruction Set

Mnemonic, Operands		Description	Cycles	14-Bit Instruction Word MSb			LSb	Status Affected	Notes
BYTE-ORIENTED FILE REGISTER OPERATIONS									
ADDWF	f, d	Add W and f	1	00	0111	dfff	ffff	C,DC,Z	1,2
ANDWF	f, d	AND W with f	1	00	0101	dfff	ffff	Z	1,2
CLRF	f	Clear f	1	00	0001	lfff	ffff	Z	2
CLRW	-	Clear W	1	00	0001	0xxx	xxxx	Z	
COMF	f, d	Complement f	1	00	1001	dfff	ffff	Z	1,2
DECF	f, d	Decrement f	1	00	0011	dfff	ffff	Z	1,2
DECFSZ	f, d	Decrement f, Skip if 0	1(2)	00	1011	dfff	ffff		1,2,3
INCF	f, d	Increment f	1	00	1010	dfff	ffff	Z	1,2
INCFSZ	f, d	Increment f, Skip if 0	1(2)	00	1111	dfff	ffff		1,2,3
IORWF	f, d	Inclusive OR W with f	1	00	0100	dfff	ffff	Z	1,2
MOVF	f, d	Move f	1	00	1000	dfff	ffff	Z	1,2
MOVWF	f	Move W to f	1	00	0000	lfff	ffff		
NOP	-	No Operation	1	00	0000	0xx0	0000		
RLF	f, d	Rotate Left f through Carry	1	00	1101	dfff	ffff	C	1,2
RRF	f, d	Rotate Right f through Carry	1	00	1100	dfff	ffff	C	1,2
SUBWF	f, d	Subtract W from f	1	00	0010	dfff	ffff	C,DC,Z	1,2
SWAPF	f, d	Swap nibbles in f	1	00	1110	dfff	ffff		1,2
XORWF	f, d	Exclusive OR W with f	1	00	0110	dfff	ffff	Z	1,2
BIT-ORIENTED FILE REGISTER OPERATIONS									
BCF	f, b	Bit Clear f	1	01	00bb	bfff	ffff		1,2
BSF	f, b	Bit Set f	1	01	01bb	bfff	ffff		1,2
BTFSC	f, b	Bit Test f, Skip if Clear	1 (2)	01	10bb	bfff	ffff		3
BTFSS	f, b	Bit Test f, Skip if Set	1 (2)	01	11bb	bfff	ffff		3
LITERAL AND CONTROL OPERATIONS									
ADDLW	k	Add literal and W	1	11	111x	kkkk	kkkk	C,DC,Z	
ANDLW	k	AND literal with W	1	11	1001	kkkk	kkkk	Z	
CALL	k	Call subroutine	2	10	0kkk	kkkk	kkkk		
CLRWDT	-	Clear Watchdog Timer	1	00	0000	0110	0100	$\overline{TO}$,$\overline{PD}$	
GOTO	k	Go to address	2	10	1kkk	kkkk	kkkk		
IORLW	k	Inclusive OR literal with W	1	11	1000	kkkk	kkkk	Z	
MOVLW	k	Move literal to W	1	11	00xx	kkkk	kkkk		
RETFIE	-	Return from interrupt	2	00	0000	0000	1001		
RETLW	k	Return with literal in W	2	11	01xx	kkkk	kkkk		
RETURN	-	Return from Subroutine	2	00	0000	0000	1000		
SLEEP	-	Go into standby mode	1	00	0000	0110	0011	$\overline{TO}$,$\overline{PD}$	
SUBLW	k	Subtract W from literal	1	11	110x	kkkk	kkkk	C,DC,Z	
XORLW	k	Exclusive OR literal with W	1	11	1010	kkkk	kkkk	Z	

Internal architecture of PIC16F84A

